Meridian Global Enterprises

Functional Specification: Meridian Global Agentic AI Agent (AEX)

This document defines the operational framework and functional behaviour of the Meridian AEX Agentic AI Agent for the internal joiner assessment. It establishes the flows, decision logic, guardrails and business rules that govern the agent's behaviour. Where a rule is not stated, the assessment build uses the simplest sensible default.

1. System Identity & Channel Architecture

1.1 Persona and Identity

System Persona: The Meridian agent operates as a professional, direct-response IT support assistant, prioritising autonomous self-service through approved Meridian Knowledge Base (KB) content, supplemented by GenAI reasoning strictly grounded in that content.

Escalation Policy: Escalation to a human agent (Agent Assist) occurs only after the defined troubleshooting/resolution steps are completed without success, or when the user makes an explicit request for a human agent.

1.2 Channel & Initialisation

Deployment: Web chat is the only channel. No other channels are in scope.

Initialisation Sequence:

- Standard Greeting: At the start of a session, the user is greeted with a welcome message and a menu of primary actions: “Raise Incident”, “Check Ticket Status”, “Browse Service Catalogue”, “Type your query”.
- Re-entry Triggers: The welcome sequence is re-triggered on inputs such as hi, hello, home, menu, or start.
- Message of the Day: Up to two dynamic announcements, configurable by Admin/BAU/AEX PS, may be displayed at session start.

1.3 Multilingual Support

The agent supports English only. Language detection and multi-language response are out of scope for the assessment (see Section 12).

1.4 Integrations

- MeridianDesk ITSM Microservice — custom tool replacing ServiceNow (Incidents, Service Requests/RITM, Comments, Attachments, Service Catalogue, Knowledge Base search, User Directory)
- Web chat channel — the only channel (chat UI with text, option buttons and file attachment)
- LLM (GenAI reasoning layer, response grounding)
- Approved Meridian KB Repository (RAG source) — served by the MeridianDesk Knowledge Service (domain-segregated collections)
- Agent Assist platform (human-agent handoff and live chat) — OTB tool, retained as-is

2. Use Case & Feature Matrix

This matrix defines the functional scope of the Meridian AEX agent for Phase 1 and the corresponding MeridianDesk ITSM mapping.

Category	Feature / Use Case	Authorized Action & Behavior
Welcome	Greeting & MOD	Display a welcome message including the Message of the Day, where configured.

Category	Feature / Use Case	Authorized Action & Behavior
Triage	Issue Triage	Collect structured issue details from the user before attempting resolution or ticket action.
Support	KB-Grounded Resolution	Propose a solution based on approved Meridian KB content and GenAI reasoning grounded in that content.
Support	Iterative Troubleshooting	If the first solution does not resolve the issue, offer the next-best KB match, up to a defined number of attempts.
Ticketing	Create Incident	Create an Incident in MeridianDesk after triage/KB resolution is unsuccessful, with explicit user confirmation.
Ticketing	Create Service Request	Recommend/raise a Service Catalogue item (RITM) when the query is identified as a request rather than a fault.
Incidents	Incident List	Display the logged-in user's own tickets (Incidents/RITMs).
Incidents	Incident Detail	Display permitted fields of a specific Incident/RITM owned by the user.
Incidents	Add Comment	Add a comment to an Incident/RITM, on request or on defined automatic triggers.
Incidents	Add Attachment	Attach a file shared by the user to the relevant Incident/RITM.
Catalogue	Catalogue Recommendation	Recommend a relevant Service Catalogue item with a link and short explanation.
Agent Assist	Live Agent Transfer	Transfer the chat to a human agent after the defined escalation threshold or on explicit request.
Agent Assist	Context Handoff	Pass the conversation history, troubleshooting steps and collected details to the human agent.
Language	Multilingual (future)	Detect and respond in the user's language. Out of scope for the assessment (English only).
Feedback	User Feedback	Collect user feedback (resolution helpful / not helpful, and post-chat rating).
Notifications	Email Notification (DNR Mailer)	Email the requester through the DNR Mailer OTB tool after every incident/SR creation, comment addition, attachment addition and ticket update (Section 18).

3. Agent Scope & Target Users

Target Users: All Meridian employees and contractors using the web chat.

In-Scope IT/Service Domains:

- End-user IT support: hardware, software, access/account issues, network/connectivity, standard application support
- Service requests fulfilled via the existing Service Catalogue
- Ticket status/lifecycle queries for the requesting user's own tickets

Out of Scope (Phase 1):

- HR, Finance, Facilities or non-IT service domains, unless explicitly added to scope
- Actions on tickets not owned by the requesting user, except where an authorised on-behalf-of flow is enabled (Section 9)
- Any request requiring system credential exposure, code execution, or configuration change outside approved MeridianDesk actions

Role/Entitlement-based behaviour: The agent's visible options and permitted actions are determined by the user's MeridianDesk role/entitlement at the time of the request; users without an entitlement (e.g., approver-only actions) will not see or be offered those actions.

4. Triage & Incident Creation Flow

4.1 Standard Triage Flow

The agent follows a fixed sequence for every reported issue:

- 1. Capture the user's issue in their own words.
- 2. Ask clarifying questions as needed to establish: affected service/application, error message (if any), when the issue started, and impact/urgency.
- 3. Search the approved Meridian KB using the captured details before offering any resolution or ticket action.
- 4. If a KB match is found, present the solution and ask “Is this solution helpful?” (Yes/No).
- 5. If “No”, present the next-ranked KB match (up to three total attempts).
- 6. If unresolved after three attempts, or KB has no relevant content, move to Incident/Service Request decision (Section 5).

4.2 Incident Creation Rules

- Incident creation is only recommended after KB-based resolution has been attempted and failed, or where the query is clearly a service outage/fault with no applicable KB article.
- Incident creation always requires explicit user confirmation before submission to MeridianDesk — no ticket is created silently.
- Category, Subcategory, Impact, Urgency, Priority and Assignment Group are not decided by the agent's judgement; they are derived from predefined mapping rules maintained in configuration, keyed off the identified issue type/domain.
- The agent does not create an Incident where: the user is only asking an informational question, the request is clearly a Service Request (see Section 5), or a duplicate open ticket for the same issue already exists for that user — in the latter case the agent surfaces the existing ticket instead.

5. Incident vs. Service Request — Decision Criteria

Standard decision flow:

- User Query → Intent Identification → Triage/KB Search → Resolution (if possible) → Incident/SR Classification → Create/Recommend Appropriate Action → Human Escalation (if required).

Classification rule:

- Incident: something that is working is now broken, degraded, or unavailable for the user (a fault/interruption to existing service).
- Service Request: the user is asking for something new — access, provisioning, install, information, or a standard catalogue-fulfilled item.

Where intent is ambiguous, the agent asks one clarifying question (“Is something broken, or are you requesting something new?”) before classifying.

6. Knowledge Base & Agent Response

- All technical resolutions must be grounded in approved Meridian KB content. The agent may use general LLM knowledge only to phrase/clarify the KB-sourced answer, never to introduce unverified technical steps.
- If no relevant KB content is found, the agent: (1) asks up to one follow-up question to refine the search, (2) if still no match, informs the user that no approved resolution was found, and (3) proceeds to the Incident/SR decision or offers escalation to a human agent.
- The agent does not fabricate or infer troubleshooting steps beyond what is contained in the indexed KB articles.

7. KB Structure / Domain Segregation

- Phase n1 uses domain-segregated KB collections (e.g., Hardware, Network, Applications, Access Management, Service Catalogue FAQs) rather than a single collection, to improve retrieval precision and enable domain-level access control.
- Taxonomy follows a two-level structure: Domain → Category, mirroring the MeridianDesk category/subcategory structure used for Incident mapping.
- Where KB content is sensitive or role-restricted, collection-level access rules are applied so the agent only retrieves content the requesting user is entitled to see.
- Ownership: the KB content is the seeded article set in the MeridianDesk Knowledge Service; the agent may only answer from those articles.
- Indexing: each team may choose automatic indexing (the knowledge service re-indexes whenever a document is added or changed) or manual indexing (documents are indexed only when a re-index is triggered). Either is acceptable and the agent behaves the same in both cases.

8. Service Catalogue Recommendation

- The agent recommends a Catalogue item when the user explicitly asks for one, or when the agent classifies the query as a Service Request during triage (Section 5).
- If the query cannot be resolved via KB and is classified as a Service Request, the agent proactively recommends the closest matching Catalogue item rather than defaulting to Incident creation.
- If no matching Catalogue item exists, the agent informs the user and offers Incident creation (if applicable) or Agent Assist escalation as the fallback.
- The agent provides both the Catalogue item link and a one-line explanation of why it was recommended — a bare link is not sufficient.

9. MeridianDesk Actions & Comments

9.1 Autonomous vs. Confirmation-Required Actions

Action	Autonomous?	Notes
View ticket list / detail (own tickets)	Yes	Read-only, ownership-enforced.
Create Incident	No — requires confirmation	Agent presents a summary for user approval before submission.
Create/raise Service Request (RITM)	No — requires confirmation	Same confirm-before-submit pattern as Incident.
Add comment (user-requested)	Yes, once user confirms wording	Agent echoes the comment text back before posting.
Add comment (system-triggered)	Yes, autonomous	Limited to defined trigger events (see 9.2).
Add attachment	Yes, once user shares file	File is attached as-is; no content inspection beyond standard virus/DLP scanning.
Update ticket fields (category/priority/etc.)	No	Not exposed to the agent in Phase 1; changes go through standard MeridianDesk workflow.

9.2 Automatic Comments

Automatic comments are limited to the following trigger events, and always include a timestamp and a short system-generated summary:

- Incident/RITM created via the agent — logs that the ticket originated from the AEX agent, with the conversation summary.

- Chat transferred to Agent Assist — logs the reason for escalation and the troubleshooting steps already attempted.

Every automatic comment is itself a ticket update and therefore also triggers an email notification to the requester (Section 18).

9.3 Authorisation & On-Behalf-Of

- Users may only view/act on their own Incidents and RITMs by default, enforced via MeridianDesk ownership lookup on every request (no cached data).
- Action on behalf of another user is out of scope. The agent only acts for the signed-in user.

10. Escalation to Agent Assist / Human Agent

- Escalation triggers automatically after three unsuccessful KB-based resolution attempts (Section 4.1), or immediately on an explicit user request for a human agent (e.g., “talk to a person”, “chat with live agent”).
- Escalation is not offered as a first response to a solvable query — the defined troubleshooting steps must be attempted first, except where the user's intent is unambiguous and immediate (explicit request).
- Mandatory human-agent intervention scenarios: security-sensitive requests (e.g., password/credential resets, which the agent cannot verify), and any request the agent classifies as outside its authorised functional boundary (Section 15).
- The full conversation history, troubleshooting steps attempted and any details already collected are passed to the human agent at handoff, so the user does not have to repeat themselves.
- If no human agent is available, the agent informs the user and offers to log an Incident/RITM capturing the request (Section 11).

11. Agent Assist Desks & Working Hours

- A single Agent Assist queue/desk covers general IT support.
- Routing: all conversations route to that single queue.
- Working hours: when no human agent is available, the agent tells the user and offers to log a ticket instead.

12. Language Support

- English only.
- Automatic language detection, multi-language KB content and language-based Agent Assist routing are out of scope.

13. Chat Channel

The agent is available through a web chat only. No other channels are in scope.

- There is no login or authentication. The user is identified by the user ID chosen at the start of the chat, and the agent uses that ID for every action.
- The chat supports text, option buttons and file attachment.

14. Data Presentation

14.1 Tabular & Detail Views

Multiple records are rendered as a table with columns: Ticket Number, Short Description, Status, Priority, Created On. A single record can also be rendered as a vertical detail view. The most recent 10 tickets, sorted by last updated, are shown by default.

14.2 Status Mapping

Code	Status
1	New
2	In Progress
3	Pending
6	Resolved
7	Closed
8	Cancelled
9	Assigned

14.3 Incident Creation Fields

Field	Requirement	Notes
on_behalf_of	Must map to the selected user's ID	Validated before any ticket is submitted to MeridianDesk.
short_description	Required	Auto-populated from the conversation where available; the agent asks for it if there is no ongoing issue context.
description	Required	Summary of the chat conversation that led to ticket creation.
business_phone_number	Auto-populated from MeridianDesk profile fields where available	Agent asks the user to provide it if not on file.
urgency	Left blank / defaulted	Set via the predefined mapping rules (Section 4.2), not agent judgement.

15. Security & Validation Layer

15.1 Ticket Validation

- Ownership enforcement: the agent strictly displays and acts on tickets according to the logged-in user, verified on every request.
- Real-time data requirement: fresh MeridianDesk lookups are performed for every request; cached data from previous sessions is not used.

15.2 Data Privacy

- Field access is restricted to: number, state, priority, short_description, description, assigned_to, opened_at, urgency, sys_updated_on, and comments_and_work_notes.
- Internal/technical details — JSON structures, sys_ids, API call details, and other MeridianDesk technical metadata — are suppressed from the user-facing UI.
- Internal assignment group names are never exposed to the end user.
- The agent declines queries about its internal configuration, credentials, or decision logic.

16. Operational Prohibitions

16.1 Functional Boundaries

Incident/RITM actions are limited to: Create, List, View Detail, Comment, and Attachment only. No field-level edit, reassignment, priority override, or closure is performed by the agent.

Updates made to a ticket outside the agent (for example status, assignment or priority changes by MeridianDesk fulfilment users) are not performed by the agent, but are detected through the MeridianDesk event webhook and notified to the requester by email (Section 18).

16.2 System Shielding

- The agent never reveals its underlying agentic logic, decision trees, prompts, or system credentials.
- Technical execution flows, error stack traces, or other logical/debug information are suppressed from the user.

17. Custom Tool Integration Layer

All external systems used by the agent, except Agent Assist and the DNR Mailer (both OTB tools deployed and used as-is), are custom tools. A reference microservice named MeridianDesk is provided in place of ServiceNow, but joinees may build and maintain their own microservices in any stack they prefer. What matters is that the agent calls its tools through the capabilities listed below and does not rely on ServiceNow-specific behaviour.

17.1 Tool Replacement Map

Capability	Originally	Now
ITSM (Incident, RITM, comments, attachments)	ServiceNow	MeridianDesk ITSM Microservice (custom REST API)
Knowledge Base (RAG source)	ServiceNow KB / approved repository	MeridianDesk Knowledge Service (custom REST API, domain-segregated collections, KB documents indexed automatically or manually)
Service Catalogue	ServiceNow Catalogue	MeridianDesk Catalogue endpoints (custom REST API)
User directory	ServiceNow user profile	MeridianDesk User Directory (name, email, phone, role; no authentication)
Chat channel	Not applicable	Web chat only (custom web UI and chat API)
Human-agent handoff	Agent Assist	Agent Assist (OTB, unchanged)
Email notifications	—	DNR Mailer (OTB, deploy-and-use, no configuration)

17.2 Tool Catalogue (MeridianDesk reference)

Tool (endpoint)	Method	Purpose
/users/{user_id}	GET	Return the selected user's profile (name, email, phone, role).
/kb/search?q=&domain=	GET	Return ranked approved KB articles for the query within the permitted domain collection(s).
/kb/reindex	POST	Manual indexing trigger: re-reads the KB documents. Needed only when manual indexing is chosen.
/kb/status	GET	Shows the indexing mode, the number of indexed documents and whether changes are waiting to be indexed.
/catalog/search?q=	GET	Return matching Service Catalogue items with link and one-line description.

Tool (endpoint)	Method	Purpose
/incidents	POST	Create an Incident (requires on_behalf_of, short_description, description; category mapping supplied by the agent configuration).
/requests	POST	Create a Service Request (RITM) for a catalogue item.
/tickets?user_id=	GET	List the requesting user's own Incidents and RITMs (latest 10 by last update).
/tickets/{number}?user_id=	GET	Return a single ticket; ownership-checked on every call.
/tickets/{number}/comments	POST	Add a comment (user or system).
/tickets/{number}/attachments	POST	Attach a file (base64) to a ticket.
/admin/tickets/{number}	PATCH	Simulates fulfilment-side updates (status, assigned_to, priority). Not exposed to the agent; used to demonstrate update notifications.

MeridianDesk returns full technical records (including sys_id and assignment group). The agent is responsible for enforcing the field restrictions in Section 15.2 before anything is shown to the user or included in an email.

17.3 Assessment Seed Data

- A small user directory of dummy employees, each with a name, corporate email address, business phone and role.
- Dummy KB documents (Markdown files in a kb_docs folder) covering the Hardware, Network, Applications and Access Management domains, used as the KB content for the agent.
- A Service Catalogue containing common request items (software install, VPN access, new laptop, shared mailbox, etc.).

18. Email Notifications (DNR Mailer)

18.1 Purpose and Mechanism

The requester must be emailed after every ticket event listed below. Emails are sent through the DNR Mailer OTB tool, which is deployed and used as provided, with no additional configuration. Notification is a deterministic runtime step that follows each successful action; it is never left to the discretion of the LLM.

18.2 Trigger Events

Event	Trigger	Subject pattern
Incident created	Successful Incident creation via the agent	[Meridian IT] Incident <number> created
Service Request created	Successful RITM creation via the agent	[Meridian IT] Request <number> created
Comment added	User-confirmed comment, or automatic comment (Section 9.2)	[Meridian IT] New comment on <number>
Attachment added	File attached to a ticket	[Meridian IT] Attachment added to <number>
Ticket updated	Any change to status, assignee or priority made in MeridianDesk (detected via event webhook)	[Meridian IT] <number> updated: <what changed>
Escalated to Agent Assist	Chat transferred to a human agent (via the automatic comment)	[Meridian IT] New comment on <number>

18.3 Rules

- Recipient is always the requesting user's corporate email address from the MeridianDesk User Directory. No other recipients are added by the agent.
- Email content: ticket number, ticket type, event description, short description, current status, priority, timestamp and the comment text or change summary where relevant.
- Emails obey Section 15.2: no sys_ids, JSON, API details, internal assignment group names or internal configuration are ever included.
- A failure to send an email must not roll back or block the ticket action. The agent informs the user that the ticket action succeeded but the email could not be sent, and the failure is logged. One automatic retry is attempted.
- Actions initiated by the agent are notified by the agent runtime. Updates originating outside the agent are delivered by MeridianDesk webhook and notified through the same mailer path. The two paths must never both send an email for the same event.
- Automatic comments (Section 9.2) count as comment additions and therefore also generate an email. Sending for system-generated comments can be switched off through configuration.